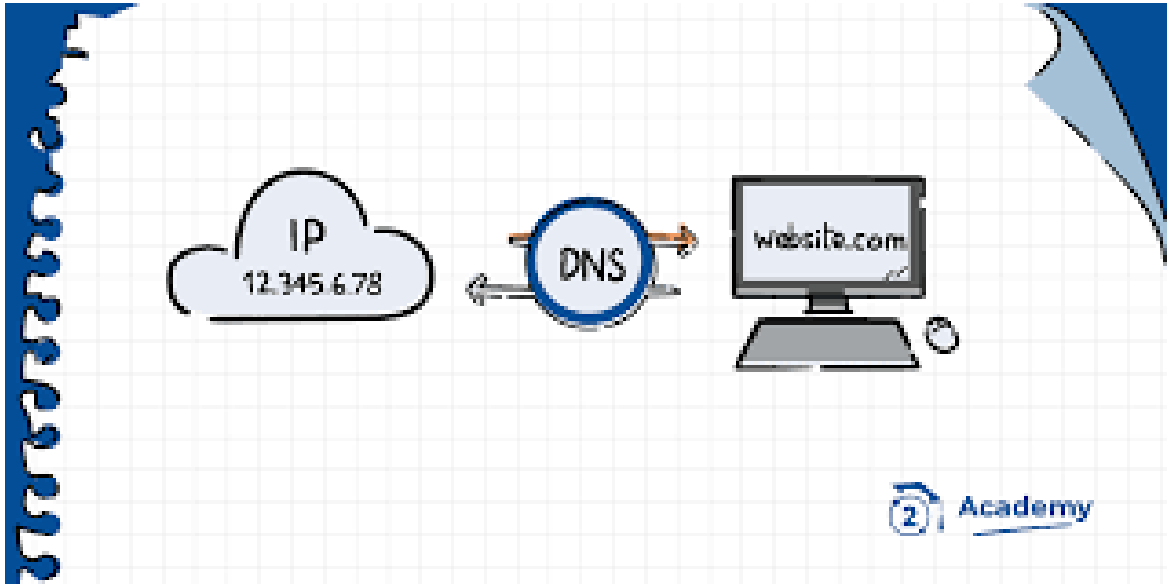
DNS



What Is DNS?

- **DNS = Domain Name System**
- Think of it as the 📖 **phonebook of the Internet**

- It **translates domain names** (like google.com) into IP addresses (like 142.250.185.206)

- Makes it easier for humans to use names instead of numbers

- Without DNS, you'd have to remember every website's IP address! 😵

Why We Need DNS

Without DNS	With DNS
You must type 142.250.185.206 to visit Google	You simply type google.com
Hard to remember and prone to mistakes	Easy to use, readable names
Every change in IP requires manual update	DNS automatically maps the new IP

How DNS Works (Step-by-Step)

1. You type google.com in your browser.
2. Your request goes to your **ISP's DNS server**.
3. If the ISP DNS knows the IP → it sends it back immediately.
4. If not → it asks other **DNS servers** for help.
5. If none know → it finally contacts the **Root DNS Server** 🌍 .
6. Once found, the ISP DNS saves the answer for next time (called caching).
7. The website loads using the correct IP address.

🧠 DNS = automatic translator between human-friendly names and machine IPs.

The DNS Layers

DNS servers are **organized like a pyramid**:

Level	Role	Example
Root Servers	Top-level servers that store all TLD (.com, .org, .uk) info	13 global root servers
TLD Servers	Manage domain extensions	.com, .org, .edu, .ae
Authoritative DNS Servers	Store specific domain info	Knows exact IP for google.com
Local/ISP DNS	Your first contact — caches previous results	Provided by Etisalat, Du, etc.

DNS in Action (Example)

- Example: You visit google.com
- Your browser → asks **Local DNS (ISP).**
- If unknown → ISP asks **TLD Server (.com).**
- If still unknown → TLD asks **Authoritative DNS.**
- That server responds with **IP address (e.g., 101.45.190.101).**
- Website loads instantly!

⚡ All this happens in milliseconds.

DNS Caching

- DNS servers and browsers store results for faster access next time.
- Example:
  - First visit to google.com → 200ms
  - Second visit → <10ms (fetched from cache)
- Saves bandwidth and reduces load on DNS servers.

🕑 Cache can expire → then DNS re-queries the address.

DNS and Web Hosting

- Your **Web Host** manages **DNS records** for your website.
- These records ensure your domain points to the correct **server IP**.
- Common DNS record types:

Record Type	Purpose	Example
A Record	Maps a domain to an IPv4 address	google.com → 101.45.190.101
AAAA Record	Maps to an IPv6 address	google.com → 2001:db8::1
MX Record	Handles email routing	mail.example.com
CNAME Record	Alias for another domain	www.google.com → google.com

Key Advantages of DNS

✅ Easy to remember domain names
✅ Automatically updates when IPs change
✅ Improves internet navigation speed
✅ Enables global access to websites

DNS Security Notes

- DNS can be attacked or misused:
  - **DNS Spoofing / Cache Poisoning**: Redirect users to fake websites.
  - **DDoS Attacks**: Overload DNS servers to cause outages.
- 💡 Use **DNSSEC** (**DNS Security Extensions**) for protection.
- Always use **trusted DNS providers** (e.g., Google DNS, Cloudflare, or your ISP).

Quick Summary

Concept	Meaning
DNS	Translates names ↔ IP addresses
Root Server	Top-level DNS authority
ISP DNS	First server to process your request
Caching	Stores results for faster access
DNSSEC	Adds security to DNS lookups